Exercise II – Object Oriented Design (Analysis)

Python Programming Assignment

1. Problem Identification

A suitable problem for an object-oriented design in the field of computer science and intelligent systems is the development of a **Sign Language Translation System**. The goal of this system is to enable communication between deaf and hearing people by translating sign language gestures into textual or spoken language.

This problem is well suited for an object-oriented approach because it involves multiple interacting entities, reusable components, and extensible functionalities.

2. Static and Dynamic Aspects

2.1 Static Aspects (Attributes)

The main object of the system can be defined as a `SignLanguageTranslator` object. Examples of its static properties (attributes) are:

- `video_stream`: the captured video input
- `frame_sequence`: extracted frames from the video
- `keypoints`: hand or body landmarks
- `language`: target translation language
- `model`: trained machine learning model

2.2 Dynamic Aspects (Methods)

The dynamic behavior of the object is represented by its methods:

- `capture_video()`: captures video data
- `extract_keypoints()`: extracts relevant features
- `predict_text()`: predicts the corresponding text
- `translate()`: performs the translation process
- `display_output()`: displays or outputs the result

3. Inheritance and Polymorphism

3.1 Inheritance

Inheritance can be used by defining a general `Translator` base class. Specialized translators inherit from this base class:

- `Translator` (base class)
- `SignLanguageTranslator` (child class)
- `SpeechTranslator` (child class)

Each child class inherits common attributes and methods while providing specific implementations.

3.2 Polymorphism

Polymorphism appears when the same method name is used with different implementations. For example, the method `translate()` can behave differently depending on the input type:

- Translating sign language from video input
- Translating speech from audio input

This allows the system to treat different translators uniformly while maintaining flexibility.

Conclusion

Object-oriented design provides a structured and scalable solution for complex systems such as sign language translation. By using encapsulation, inheritance, and polymorphism, the system becomes easier to maintain, extend, and reuse.